

Here is the Detailed Design Document (DDD) for establishing a **Holonic Semantic Architecture** within Baxter’s R&D, Clinical Trials, and Manufacturing operations.

This document defines the architectural blueprint for transitioning legacy, siloed data into a unified, AI-ready digital thread, ensuring emergent capabilities are strictly governed by top-down semantic models.

Detailed Design Document: Holonic Semantic Architecture for R&D and Tech Transfer

1. Executive Summary

This design establishes a scalable, governed, and modular architecture designed to harmonize data across disparate systems, labs, and clinical trial repositories. By utilizing a "Semantic-First" approach combined with an enterprise low-code orchestration layer, this architecture enables the safe deployment of AI agents while ensuring strict GxP compliance, data literacy, and seamless tech transfer from R&D into existing and future manufacturing sites.

2. Tool Alternatives & Recommendations

To realize this architecture, specific technology stacks were evaluated against the need for rigorous semantic governance, physical equipment integration, and enterprise application scalability.

	Siemens Mendix (Recommended)	Stardog / GraphDB	Neonto Editor
Category	Category Execution & Orchestration	Category Knowledge Graph & Semantic Storage	Category Semantic Lifecycle Governance
Alternatives Evaluated	Alternatives Evaluated	Alternatives Evaluated	Alternatives Evaluated

	Decisions + ProcessMaker, Red Hat PAM	AWS Neptune, Neo4j	TopBraid EDG, Protege
Rationale	Rationale Native Siemens Xcelerator integration (Teamcenter/Opcenter) is critical for hardware/manufacturing tech transfer. Superior multi-experience UI for scientists.	Rationale Requires native support for RDF/OWL reasoning, SWRL, and SHACL validation, which property graphs (Neo4j) lack out-of-the-box.	Rationale Balances visual lifecycle management with robust underlying OWL/SHACL support, allowing business users to manage rules without writing code.
Role in Architecture	Role in Architecture Holon UI, Orchestration, Human-in-the-Loop workflows.	Role in Architecture Single source of truth for all data, metadata, and relationships.	Role in Architecture Authoring environment for ontologies, SHACL constraints, and SWRL analytics.

3. The 7-Layer Holonic Architecture

This updated 7-layer model maps Baxter's physical reality to its digital governance structures, enabling modular control.

Layer 7: Enterprise Ecosystem (The Consumers)

- **Systems:** SAP (ERP), Siemens Teamcenter (PLM), Veeva (Regulatory/Clinical QMS).
- **Function:** Consumes the "finished" data packages (tech transfers, clinical batch requirements, labeling data) produced by the lower layers.

Layer 6: Holon Orchestration & UI (The Interface)

- **Tool:** Siemens Mendix (Control Center, Data Hub).

- **Function:** The primary interaction layer. Provides scientists, lab managers, and clinical operators with custom, role-based dashboards. Orchestrates workflows across domains and handles human approvals.

Layer 5: Semantic Governance (The Rulebook)

- **Tool: Neonto Editor** (Managing SHACL/SWRL).
- **Function:** The central repository for all business logic, regulatory constraints, and data definitions. Defines the boundaries of what is "legal" within any given Holon.

Layer 4: Knowledge Graph (The Digital Thread)

- **Tool: Stardog / GraphDB.**
- **Function:** Stores the harmonized data as RDF instances. Virtualizes siloed legacy databases (LIMS, ELNs, Clinical Repositories) into a single, queryable semantic network.

Layer 3: Policy Enforcement Point (PEP) & AI Edge

- **Tool: Siemens Industrial Edge** (containing Python/SHACL validation engines & AI Agents).
- **Function:** The gateway. All system-to-system or AI-generated process changes must pass through this layer to be validated against the Layer 5 rulebook before reaching physical equipment.

Layer 2: Digital Twin & Connectivity

- **Tool: Siemens MindSphere / OPC-UA / Lab APIs.**
- **Function:** The translation layer. Converts raw machine telemetry (temperature, pressure, pH) and legacy API responses into RDF triples to feed the Knowledge Graph.

Layer 1: Physical Holons

- **Assets:** Bioreactors, automated pipettes, lab robots, raw clinical materials, and environmental sensors.
- **Function:** The physical execution of R&D processes.

4. Components: Custom vs. Out-of-the-Box (OOTB)

To support the architecture, the **Siemens Mendix** footprint will utilize a blend of standard and purpose-built components.

- **OOTB Mendix Components:**
 - **Data Hub / OData Connectors:** To virtualize ERP/PLM data without replicating it into the R&D space.
 - **Workflow Engine:** Standardizes task routing and multi-step approval processes for GxP sign-offs.

- **AI Agent Task Widget:** Monitors the active status and decision trees of deployed AI agents.
- **Custom React Components (Built in Mendix Studio Pro):**
 - **SHACL Validation Viewer:** Translates raw SHACL validation reports into a traffic-light UI (Green/Yellow/Red) for R&D managers to easily spot compliance violations.
 - **Ontology Context Explorer:** A visual "Tree Node" widget allowing scientists to explore how a specific clinical material relates to downstream supply chain elements.
 - **HITL Intervention Pane:** A side-by-side comparison tool used when an AI proposes a change, showing "Current State" vs. "Proposed AI State" with highlighted impacts.

5. Governance, Constraints, and Human-in-the-Loop (HITL)

Governance in this architecture is dynamic. It is enforced at the moment of action via the **Policy Enforcement Point (PEP)**.

1. **Rule Authoring (Neonto):** A Quality Manager defines a SHACL shape: *"A Clinical Batch must have a recorded pH between 7.0 and 7.4 before moving to packaging."*
2. **Emergent AI Action:** An AI agent optimizing the line suggests altering the buffer concentration, which predicts a pH of 7.5 to save 10 minutes of processing time.
3. **PEP Intercept:** The AI's proposed RDF payload is intercepted at Layer 3 and checked against the Knowledge Graph (Layer 4) and SHACL constraints.
4. **HITL Trigger:** The SHACL engine flags a "Violation." The PEP blocks the automated change and triggers a Mendix Workflow.
5. **Resolution:** An R&D scientist receives an alert on their Mendix dashboard, reviews the AI's proposal via the HITL Intervention Pane, and either rejects the change or overrides it with a logged justification (satisfying 21 CFR Part 11 requirements).

6. Operating Flows: R&D to Tech Transfer

Flow A: Dynamic Lab Execution (Emergent Capabilities)

- **Context:** A robot is executing a new mixing protocol for an experimental formulation.
- **Action:** The robot uses local machine learning to adjust its impeller speed based on real-time viscosity data (Layer 1 -> Layer 2).
- **Validation:** Before publishing this "new best practice" to the broader lab, the robot packages the parameter change as an RDF event. The Edge PEP (Layer 3) validates it against safety SHACLs.
- **Publishing:** If valid, the new process is written to the Knowledge Graph (Layer 4) and becomes immediately available to other lab robots as a permitted emergent behavior.

Flow B: Tech Transfer to Manufacturing (Scaling & Mapping)

- **Context:** R&D has finalized a drug formulation and clinical trial materials are ready for scale-up.
- **Action:** A scientist uses the Mendix Tech Transfer Dashboard to initiate handover.
- **Translation:** Because R&D data was natively structured as RDF against the top-down ontology, the system automatically runs SWRL rules to map "R&D Lab Parameters" to "Manufacturing Line Constraints."
- **Integration:** Mendix pushes this harmonized data package directly into Siemens Teamcenter (PLM) as a unified Bill of Process (BOP) and updates SAP (ERP) with the required raw material BOMs, completing the digital thread from discovery to production.